SE444: Introduction to Artificial Intelligence
Makeup Mid-Term Exam I — Fall 2025

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Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Clearly state any assumptions made that might be needed to understand your solution.
- Show your work – any sign of serious effort will be considered.
- No marks will be given for not making any effort.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	Mapped CLOs
Part I MCQ	A* Heuristics & Optimality	5 min	1		CLO 1
	Alpha-Beta Pruning Analysis	5 min	1		CLO 1
	Exploration vs. Exploitation	5 min	1		CLO 1, 2
	Local Optima & Escape Mechanisms	5 min	1		CLO 1, 3
Part II A*	Problem Formulation	5 min	1		CLO 1
	Trace A* Search	15 min	3		CLO 1, 2
	Comparative Analysis	10 min	2		CLO 1
Part III GA	Chromosome Representation	5 min	1		CLO 2
	Fitness Computation	10 min	2		CLO 2
	GA Operators & Analysis	15 min	2		CLO 2, 3
	Total for Part I (MCQ)	20 min	4		
	Total for Part II (A* Search)	30 min	6		
	Total for Part III (GA)	30 min	5		
	Total Points	80 min	15		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Question I.1: A* Search — Heuristic Design and Optimality Guarantees

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are designing an A* search agent for a package delivery robot navigating a warehouse grid. The robot can move in four cardinal directions (N, S, E, W) with cost 1, or diagonally with cost $\sqrt{2}$. You consider three candidate heuristics for estimating the remaining cost to the goal:

- $h_1(n)$ = Euclidean distance from n to goal
- $h_2(n)$ = Manhattan distance from n to goal
- $h_3(n) = \max(h_1(n), h_2(n))$

Evaluate each of the following statements and select **all** that are correct.

A) h_1 is admissible because the straight-line distance never overestimates the actual shortest-path cost when diagonal moves are allowed.

B) h_2 is always admissible for this problem regardless of the allowed movement types.

C) h_3 dominates both h_1 and h_2, making A* with h_3 explore fewer nodes than A* with either h_1 or h_2 alone.

D) If we multiply h_1 by a factor of 1.5 to create $h_4(n) = 1.5 \cdot h_1(n)$, A* with h_4 may find suboptimal solutions but will typically run faster.

E) If the robot's diagonal movement cost is changed from $\sqrt{2}$ to 1.0, then h_1 becomes inadmissible.

Question I.2: Alpha-Beta Pruning — Subtree Analysis

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a minimax game tree where MAX plays first. During alpha-beta search with initial $\alpha = -\infty$ and $\beta = +\infty$, the algorithm explores nodes left-to-right. At some MIN node M, after evaluating its first child, the current values are $\alpha = 7$ (inherited from parent) and $\beta = 5$ (computed from the child just evaluated).

Which of the following statements correctly describe what happens next?

A) The remaining children of M can be pruned because $\alpha \geq \beta$, meaning MAX already has a better guaranteed option.

B) The value returned by node M to its parent MAX node will be exactly 5.

C) If the pruned subtrees contained a terminal node with value 3, including it would have changed the minimax value at the root.

D) The pruning is called "beta cutoff" because we are pruning at a MIN node where the beta bound was exceeded.

E) If we had searched the children of M in a different order, we might have explored more nodes before discovering the cutoff condition.

Question I.3: Exploration vs. Exploitation — Algorithmic Trade-offs

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

A robot learning to navigate a maze uses different strategies that balance exploration (trying new paths) and exploitation (following known good paths). Analyze the following statements about various algorithms and their exploration-exploitation characteristics.

Select all correct statements.

A) In simulated annealing, maintaining a high temperature for too long improves exploration but can prevent convergence to an optimal solution.

B) A genetic algorithm with 100% elitism (all parents copy to the next generation) eliminates both exploration and exploitation, causing the population to stagnate.

C) In simulated annealing, setting the initial temperature T_0 equal to the maximum possible energy change $\Delta E_{\max}$ ensures that nearly all moves are initially accepted, maximizing early exploration.

D) Hill-climbing with random restarts balances exploration and exploitation: each restart explores a new basin, while the climb within each basin exploits local structure.

E) In A* search, the heuristic $h(n)$ drives exploitation of promising regions, while the path cost $g(n)$ enforces exploration of low-cost paths.

Question I.4: Local Optima — Escape Mechanisms in Optimization Algorithms

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

Local search algorithms often get trapped in local optima. Different algorithms employ various mechanisms to escape or mitigate these traps. Analyze the following statements about these mechanisms.

Select all that are correct.

A) Hill-climbing inevitably stalls on plateaus or ridges because there is no way to make progress without gradient information, so such situations cannot be mitigated.

B) In genetic algorithms, the crossover operator can escape local optima by combining genetic material from two parents that are trapped in different local optima, potentially producing offspring in a better region.

C) Stochastic hill-climbing, which accepts any improving move with probability proportional to the improvement, is guaranteed to eventually find the global optimum given infinite time.

D) Simulated annealing with a sufficiently slow cooling schedule has a theoretical guarantee of converging to the global optimum, but this guarantee requires impractically long runtimes.

E) Random-restart hill-climbing is always superior to standard hill-climbing because multiple restarts guarantee finding the global optimum.

Part II. A* Search on Sliding Puzzle

Total Score: 6 points **Total Time:** 30 minutes

Context: You are solving a 3-puzzle (2×2 sliding tile puzzle) using A* search. The puzzle has tiles numbered 1, 2, 3, and one blank space (denoted by 0). The goal is to arrange tiles in the following configuration:

Goal State:

1	2
3	0

We represent states as row-major lists: [top-left, top-right, bottom-left, bottom-right]. Thus, the goal state is represented as: `[1, 2, 3, 0]`

Operators: The blank (0) can swap with any adjacent tile (up, down, left, right) if the move is valid within the 2×2 grid.

Path cost: Each move costs 1.

Heuristic $h(n)$: Number of misplaced tiles (not counting the blank).

Question II.1: Problem Formulation

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Formulate the 3-puzzle as a search problem. Clearly define:

1. **State space**: What constitutes a valid state?
2. **Initial state**: $S_0 = [3, 1, 0, 2]$
3. **Actions**: What moves are available from any state?
4. **Transition model**: How does an action transform the state?
5. **Path cost function**: How is the cost of a solution measured?
6. **Goal test**: How do we know we've reached the goal?

Question II.2: Trace A* Search

Time: 15 minutes **Score:** 3 points **Mapped CLOs:** CLO 1, CLO 2

Starting from $S_0 = [3, 1, 0, 2]$, trace A* search using $f(n) = g(n) + h(n)$.

Instructions:

- For each expansion, show:
  1. The current state being expanded
  2. Its $g(n)$, $h(n)$, and $f(n)$ values
  3. All generated successors with their f-values
- Update the frontier (priority queue) after each expansion
- **Tie-breaking rule:** When f-values are equal, expand the state that comes first in lexicographic order (e.g., [0,1,2,3] before [0,1,3,2])
- Continue until you reach the goal state
- Show the complete solution path and its total cost

Hint: Draw the 2×2 grid for each state to visualize tile positions and valid moves.

Initial State $S_0 = [3, 1, 0, 2]$**:**

3	1
0	2

Goal State = [1, 2, 3, 0]:

1	2
3	0

Question II.3: Comparative Analysis

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 1

Answer the following questions based on your A* trace:

(a) **(1 point)** Consider an alternative heuristic $h'(n)$ = Manhattan distance (sum of horizontal and vertical distances of each tile from its goal position).

- Is h' admissible for this problem? Justify your answer.
- Which heuristic (h or h') would you expect to expand fewer nodes? Explain why.

(b) **(1 point)** If we used Greedy Best-First Search (using only $h(n)$, ignoring $g(n)$) instead of A*:

- Would the solution found necessarily be optimal?
- Describe a scenario where Greedy might expand fewer nodes than A*, and another where it might expand more.

Part III. Genetic Algorithm for Exam Scheduling

Total Score: 5 points **Total Time:** 30 minutes

Context: You are designing a Genetic Algorithm (GA) to schedule 4 final exams across 3 time slots and 2 examination halls.

Each chromosome represents a schedule for all 4 exams.
It has 4 genes, one for each exam, in the order: `[E1, E2, E3, E4]`
Each gene encodes the Timeslot–Hall assignment (e.g., `T2H1` means Timeslot 2, Hall 1).

Exam Information

Exam	Enrolled Students	Course Instructor
E1 (Calculus)	80	Dr. Smith
E2 (Physics)	60	Dr. Smith
E3 (Chemistry)	45	Dr. Jones
E4 (Biology)	50	Dr. Jones

Shared Students (students enrolled in multiple courses)

Exam Pair	Shared Students
E1 (Calculus) & E2 (Physics)	25
E2 (Physics) & E3 (Chemistry)	15
E3 (Chemistry) & E4 (Biology)	30
E1 (Calculus) & E4 (Biology)	10

Resources

Resource	Capacity / Availability
Hall H1	100 students
Hall H2	70 students
Timeslots	T1 (Morning), T2 (Afternoon), T3 (Evening)
Dr. Smith	Not available at T3

Constraints and Penalties

Constraint	Penalty
Hard: Hall-time clash — Two exams in the same hall and timeslot	+100
Hard: Instructor clash — Same instructor's exams at the same time	+100
Hard: Instructor availability — Dr. Smith's exams scheduled at T3	+100
Hard: Hall capacity — Enrolled students exceed hall capacity	+100
Soft: Student conflict — Shared students have exams at the same time	+20 per conflict

Fitness Function: Total penalty $= 100 \times$ (hard violations) $+ 20 \times$ (student conflicts)
Goal: Minimize total penalty (lower = better schedule)

Question III.1: Chromosome Representation

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 2

(a) Show how the following schedule is represented as a chromosome:
E1 → T1H1, E2 → T2H1, E3 → T1H2, E4 → T3H1

(b) How many total possible chromosomes exist in this search space? Show your calculation.

(c) Explain one advantage and one disadvantage of this representation compared to a binary encoding.

Question III.2: Fitness Computation

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 2

Given the following two chromosomes:

- $S_1 =$ [T1H1, T1H2, T2H1, T2H2] (E1→T1H1, E2→T1H2, E3→T2H1, E4→T2H2)
- $S_2 =$ [T1H1, T2H1, T1H1, T3H2] (E1→T1H1, E2→T2H1, E3→T1H1, E4→T3H2)

For **each** chromosome:

(a) Identify all hard constraint violations and count them.

(b) Identify all soft constraint violations (student conflicts at same time) and count them.

(c) Compute the total penalty.

(d) Which chromosome is fitter? Explain.

Question III.3: GA Operators and Critical Analysis

Time: 15 minutes **Score:** 2 points **Mapped CLOs:** CLO 2, CLO 3

Part (a): Selection Pressure Analysis (0.5 points)
Consider two selection strategies:

- **Strategy A:** Tournament selection with tournament size 2
- **Strategy B:** Tournament selection with tournament size 5

Which strategy has higher selection pressure? Explain how this affects the exploration-exploitation balance and the risk of premature convergence.

Part (b): Crossover Operation (1 point)
Two parent schedules are given:

- Parent 1: [T1H1, T1H2, T2H1, T2H2]
- Parent 2: [T2H1, T3H1, T1H2, T2H1]

(i) Perform **two-point crossover** with crossover points after positions 1 and 3 (so genes at positions 2 and 3 are swapped between parents).

(ii) For **one** of the resulting offspring, compute the total penalty.

(iii) Did the crossover produce a child better than both parents, worse than both, or in between? What does this illustrate about the GA search process?

Part (c): Mutation Analysis (0.5 points)
Suggest a mutation operator suitable for this problem. Then answer:

- If the mutation rate is set too high (e.g., 50% of genes mutated per generation), what problem might arise?
- If the mutation rate is set too low (e.g., 0.1%), what problem might arise?
- What mutation rate would you recommend, and why?